



PROFILE

Senior Engineering student majoring in Computer Systems at Da Nang University of Science and Technology (DUT). Possessing a solid technical foundation in embedded systems (STM32, ESP32), electronic circuit design (Altium, Proteus), and AI integration. Seeking a professional internship opportunity to apply technical analytical thinking and hardware troubleshooting skills to optimize testing processes and contribute to high-tech product development

CONTACT



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TECHNICAL SKILLS

- **Programming Languages:** C/C++ (OOP, Pointer, STL, Data Structures & Algorithms), Python, MATLAB.
- **Software & Tools:** Altium Designer, Proteus, MATLAB/Simulink, STM32CubeIDE, Keil C, Visual Studio Code, PyCharm
- **Embedded Systems:** STM32, ESP32, Arduino
- **Core Competencies:** PCB Design, Hardware Troubleshooting, Object-Oriented Programming (OOP), Image Processing (OpenCV), IoT Systems

LANGUAGES

English
Japanese

CAO TIEN DUNG

EMBEDDED PROGRAMING

HIGHLIGHTED PROJECTS

Tourism Management & Shopee Simulation | Dự án năm 1 2022 – 2023

- **Technologies:** C++, OOP (Class), Vector, Pointer, File handling.
- **Results:** Mastered object-oriented programming logic and memory management to build a backend system for managing users, orders, and product data.

OCL Audio Power Amplifier Circuit Design | Đồ án Mạch Điện Tử 2024 – 2025

- **Technologies:** Altium Designer, Proteus, Soldering, Discrete Components
- **Description:** Calculated technical parameters, designed schematics, and laid out the PCB for an OCL circuit.
- **Results:** Optimized the design via Proteus simulation and implemented the physical circuit, ensuring stable output power with low signal distortion.

FireGuard IoT - AI Smart Fire Alarm System 2025- 2026

- **Technologies:** Python (YOLOv8, OpenCV), ESP32, Streamlit, Telegram API.
- **Description:** Built an early warning fire detection system using YOLOv8, reducing the false alarm rate to under 10% through a 10-frame noise filtering algorithm.
- **Results:** Successfully deployed an automated alert system: auto-triggering hardware alarms and sending real-time field images via Telegram.

AI-based Automatic Sorting Conveyor System (PBL4)

- **Technologies:** AI / Computer Vision, C/C++, OpenCV, STM32/ESP32, Python
- **Description:** Developed a central control module using C++ (OOP) on STM32; optimized algorithms for product coordination and precise sorting via Motor Driver.
- **Technical details:** Integrated an AI-based camera system and OpenCV for real-time barcode/QR code scanning; built a database management system and automated QR payment integration.

EDUCATION

Da Nang University of Science and Technology (DUT) | 2022 - Present

Major: Computer System (5 Year-Engineering Program)

Faculty: Electronics and Telecommunications

GPA: 3.15/4.0

LANGUAGES: Japanese: JLPT N4 (Pursuing N3)

English: TOEIC 650 (Intermediate)

CERTIFICATIONS

C/C++ Programming Certificate – Issued by Codelearn

Python Programming Certificate – Issued by Coursera